

Rwanda Mathematical olympiad

IMO MOCK TEST 2

July 10, 2025

INSTRUCTIONS

- Each problem is worth 7 points.
- Only Full proofs deserves all the 7 points.
- Use only blue or black ink pens.
- Pencils can be used.
- Straight rulers, compass and colors can be used but the protractor is not allowed.
- Start each question on a new paper.

Problems

1. A grasshopper at a point (x, y) in the coordinate plane moves in the following manner. If $\min\{x, y\} \leq 0$, then the grasshopper stops moving. Otherwise, the grasshopper moves to one of the points $(x, \frac{x(3x-y)}{x+y})$ or $(\frac{y(3y-x)}{x+y}, y)$.
 - (a) Suppose that the grasshopper started at the point $(21, 28)$. Show that it always lands on an integer point regardless of how many moves it makes.
 - (b) Can the grasshopper reach point $(2025, 2026)$ starting from the point $(21, 22)$ by repeating the move?
2. Let ABC be a general triangle with orthocenter H . Denote by E_a the midpoint of segment AH and by H_a the projection of point H onto side BC ; analogously introduce points E_b, H_b, E_c, H_c . Also denote the intersections $\{A_1\} = E_bH_c \cap E_cH_b$, $\{B_1\} = E_cH_a \cap E_aH_c$, $\{C_1\} = E_aH_b \cap E_bH_a$. Show that lines AA_1, BB_1, CC_1 are concurrent.
3. Find all pairs a, b of positive integers such that for all sufficiently large positive integers n , each of the numbers $(a^n + 1)(b^n + 1)$ and $((a + b)^n + 1)((a - b)^n + 1)$ can be written as the sum of squares of two integers.